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"The Role of Medicine: Dream, Mirage, or Nemesis?" Thomas McKeown, The Nuffield Provincial Hospitals Trust, 3 Prince Albert Road, London N.W.1, 1976. £3.25.: [Introduction]

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The Role of Medicine: Dream, Mirage, or Nemesis? Thomas McKeown, The Nuffield Provincial Hospitals Trust, 3 Prince Albert Road, London N.W.1, 1976. £3.25.

The dual aims of this book are set forth in its opening paragraph as "(a) to examine the validity of a concept which is rarely stated explicitly but on which medical activities largely rest, namely that human health depends essentially on a mechanistic approach based on understanding of the structure and function of the body and of the disease processes that affect it; and (b) to consider the significance of the conclusions for medicine, particularly in relation to health services, medical education, and medical research."

The first part of the book thus addresses the determinants of health and the second, the role of medicine.

A disease classification is introduced which is followed in much of the subsequent discussion. One category includes diseases which are determined at fertilization and this is further divided into genetic diseases and "conditions associated with the genetically programmed wearing out of organs at the end of life." The second category comprises conditions which occur only in an appropriate environment. This class is subdivided according to the prenatal or postnatal effect of the environmental influences.

Mortality in England and Wales has decreased from over 20 deaths per 1,000 population in 1851 to fewer than 8 deaths per 1,000 in 1971. Almost three-quarters of the decrease in the period 1851–1971 is attributed to a fall in mortality from conditions associated with micro-organisms. The largest portion of this change (40 percent) was in mortality from airborne diseases, most conspicuously respiratory tuberculosis which accounted for 17 percent of the total decline. Twenty-one percent of the decrease was from waterborne and food-borne diseases, primarily cholera, diarrhea, and dysentery. The remaining decrease, 13 percent of the total, is accounted for among other diseases attributable to micro-organisms, such as those within the rubric convulsions and teething. Counter to

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the usual trend of declining mortality during the twelve decades under study was the increase in the death rates for rheumatic heart disease, other cardiovascular disease, and cancer.

Four causes for the decline in mortality from infectious diseases are advanced and assessed. These are: 1. changes in the character of the diseases, i.e., changes in the interaction between organism and host; 2. developments in immunization and therapy; 3. reduction of exposure to infection; and 4. improvements in nutrition. Examination of the probable effect of each of these indicates that the changes in mortality may be attributed primarily to improved nutrition. Reduced exposure to infection had some effect upon mortality during the nineteenth century, as did immunization and therapy in the twentieth. There is little evidence of general changes in the character of the infectious diseases during the period under study.

About one-quarter of the decrease in mortality in England and Wales between 1851 and 1971 was a result of changing death rates for non-infective conditions. Deaths from infanticide and starvation may have been decreasing during the nineteenth century, but most of the contribution to a more favorable mortality experience from non-infectious conditions occurred after 1900. This category of diseases is a heterogeneous one and has been subject to changes in terminology and classification. The conditions accounting for most of the mortality change within this group are those associated with prematurity and old age.

Identification of improvement in nutrition as the major causal factor in the decline in mortality is based upon the assumption of a large increase in food supplies after 1800 and the postulation of a nutritional improvement as a necessary condition for both the observed population growth and mortality decline. The decrease in mortality from water-borne and food-borne diseases appears to be related to the various improvements in hygiene which occurred during the latter half of the nineteenth century.

A number of infectious diseases have been subject to notable advances in immunization and therapy. Their mortality was decreasing prior to the introduction of these medical advances which during this century had little observable impact upon the continuing downward trends. The conspicuous exceptions to this general conclusion are the developments of streptomycin therapy for tuberculosis and poliomyelitis vaccine.

The significant change in reproductive behavior, which resulted

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in a declining birth-rate, has been of major importance. A restriction in numbers was a prerequisite to the realization of the benefits from the other factors inducing a declining mortality.

Certain residual health problems remain as mortality from infectious diseases has been reduced. These include the congential diseases and disabilities, psychiatric illness, and a group of other diseases which may be preventable. Progress toward their prevention and control is likely to be more dependent upon modification of behavioral and environmental influences than upon medical interventions.

The second section of the book, on the role of medicine, begins with a discussion of the implications of the observations and conclusions from the first section for present-day health services. These findings suggest initiatives for public policy relating to nutrition, behavior, and the environment. In the provision of personal health services, the appraisal of the effectiveness of medical technology indicates a need to redress the balance between technology and care. It is essential that a more critical stance be adopted toward quality issues in health services and that adequate evaluations be conducted prior to the general introduction of new procedures.

The scope of medical education should be expanded to place increased emphasis upon the environmental and behavioral influences acting upon a nation's health and upon an expanded and integrated concept of personal care through all stages of illness. These changes may have some effect upon the selection of students entering upon medical training but their most important impact will be upon the medical curriculum. The student should be trained to understand better the origins of disease, the effectiveness of medical interventions and how they should be assessed, and the needs of the patient beyond the acute episode of illness. The medical school and its hospital must cease their current isolation from many of the major health problems; they must also expand their interests and services to expose students to the total range of illness, disability, and modalities of care.

Advances in health have been based upon changes in nutrition, hygiene, reproductive behavior, immunization, and therapy. Some of these have been derived from medicine but some of the most important have resulted from developments in other areas of science and technology. Without abandoning the laboratory sciences, medi-

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cal research should give greater attention to health intelligence and epidemiology. The congenital abnormalities and mental illness, in particular, offer a serious and perhaps intractable challenge to our understanding of their etiology and control.

The final chapter of the book reflects upon recent critiques of medicine which view it as a dream, a mirage, or Nemesis. The dream of medical progress may be fading and the limits of laboratory medicine may have been reached. The mirage of medicine is the elimination of disease. Health problems continue, and as the diseases associated with poverty are controlled they will be replaced by those associated with wealth. The interpretation of medicine as Nemesis views it as harmful and restrictive of indivdual freedom. The author states his view of the role of medicine to be "To assist us to come safely into the world and comfortably out of it, and during life to protect the well and care for the sick and disabled."

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